Assessment Task 3: Professional Learning Report

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1. Introduction

The following report reflects my professional development throughout the semester in accordance with the Professional Learning Plan (PLP) I developed and set out for myself in Task 1. The purpose of the plan was to identify gaps existing within my current knowledge base and allow me to develop the technical skills required to bridge these gaps and deepen my professional understanding required for my future career in the computer science and information technology industry.

The PLP established several professional development goals intended to broaden my existing knowledge, addressing areas which I have identified potential for improvement. These goals cover both professional awareness and technical skills, recognising careers are more than just simply possessing programming skills, focusing on developing an understanding of ethical principles and practices within a professional context in relation to the IT industry, especially relating to Indigenous communities, building foundational knowledge in the area of cybersecurity and the C++ language, and building upon my existing Godot knowledge base to develop foundational skills and a familarity with GDScript and the engine as a whole.

These goals were selected to complement my existing studies in which I am currently undertaking a Bachelor of Computing Science (Honours) majoring in Enterprise Software Development, while also providing me with the opportunity to explore areas beyond the scope of my predefined coursework. While my studies have provided me with a foundational understanding of many core skills in programming, networking, and in software development, I recognise the important of continuing to independently develop my skills, taking responsibility for my own learning. The assigned activities compel me to engage in structured learning resources outside of coursework, completing practical exercises and reading documentation, before applying newly acquired knowledge practically to strengthen technical capabilities and become more confident with unfamiliar concepts.

The following report examines the rationale behind the professional development activities defined in my PLP, evaluating the extent to which they have contributed to the completion of my overall learning objectives. It presents evidence of achievement for each goal, summarising what each activity produced and providing all supporting artefacts, highlighting the knowledge and skills of which I have developed alongside areas where further learning may still be necessary. Furthermore, the report reflects upon my overall professional learning experience throughout the semester, analysing the effectiveness of the approach, the strengths and limitations of my learning process, and the extent of all progress made towards the completion of my goals established in Task 1, before informing of the development of further learning goals aimed towards

expanding upon my gained skillset and knowledge base throughout this task and preparing me for the challenges and expectations of my future career.

2. Justification of professional development activities

The PLP identified four required and professional skill gaps in my existing knowledge and skills, each goal selected to address a specific area for improvement while complementing my existing studies. These goals focused on developing and understanding ethical principles and practices within the IT industry, particularly in relation to Indigenous communities; building foundational knowledge in cybersecurity; gaining practical experience and developing my foundational skills with the Godot engine; and developing introductory C++ programming skills.

The activities proposed for each goal were based on their relevance to my learning objectives and potential to support independent learning, focusing on engaging in structured learning materials, online tutorials, and practical exercises. Through this I aimed to develop a stronger theoretical understanding whilst applying my newly acquired skills. The following sections explain the rationale behind each goal and how the proposed activities contribute to my professional development.

2.1. Goal 1 / Demonstrate understanding of ethical principles and practices in a professional context, especially in relation to Indigenous Communities

Developing an understanding of ethical principles and practices is essential for working responsibility within a professional context (Australian Computer Society, 2023). Computing professionals must consider the broader implications of their work, especially in relation to how their technological solutions affect individuals and communities. This is particularly important when regarding Indigenous communities, where cultural considerations and the needs of individuals and the community should inform the development and implementation of technological solutions (Australian Institute of Aboriginal and Torres Strait Islander Studies, 2020).

Through my studies, my coursework, particularly the subject Communications for IT Professionals, has provided a foundation for the completion of this goal. In this subject, I developed a technological solution intended to address the needs of the Indigenous community of Saibai Island, requiring much consideration of the community's unique individual circumstances and cultural context, allowing me to develop a more comprehensive understanding of ethical principles and practices in regard to the consideration of Indigenous communities, however, my knowledge is still lacking.

To address this gap in my knowledge, my PLP proposed the completion of the learning modules *Working Ethically in the IT Industry* and *Indigenous Professional Capability*.

These activities were selected as they are both easy to access, being provided by the subject and available readily on Canvas and directly address the areas identified for improvement. By completing these modules, I intend to build upon my previous academic experience gained through my coursework, strengthening both my understanding of ethical principles and practices, and develop a greater awareness of the considerations involved in working respectfully with Indigenous communities.

2.2. Goal 2 / Demonstrate foundational skills in Cybersecurity

My previous studies and coursework, particularly in Network Fundamentals and Web Systems, introduced me to the technical foundations underlying the field of cybersecurity. These subjects allowed me to develop a foundational understanding of how computer systems communicate and operate, specifically in relation to Unix systems and network protocols. As developing this knowledge is particularly relevant to my major of Enterprise Software Development, however, I believe it important to explore this area in greater depth, forming a deeper knowledge base, as cybersecurity is to be considered throughout ever software development process and not as an afterthought.

To build upon this existing foundation, I included the completion of the *Introduction to Cybersecurity* module in my PLP. As the resource is readily available and provided by the subject, providing an accessible way to explore cybersecurity beyond the scope of my previous coursework. Completing the module would allow me to develop a broader understanding on common threats faced and the practices implemented to mitigate them, threatening my ability to recognise and understand the importance of cybersecurity when developing and maintaining software.

Establishing these foundational skills would provide the basis for additional further learning, complimenting my existing technical knowledge. Although this activity was not intended to develop an advanced knowledge base, it would allow me to broaden my understanding of security threats, maintaining system integrity, and reducing vulnerabilities. This advanced knowledge base would support my ongoing development as a computing student, helping to prepare me to consider cybersecurity more effectively in future development projects.

2.3. Goal 3 / Develop foundational skills within the Godot Engine

As an activity to broaden my programming experience and explore the field of games development, I opted to learn Godot Engine. Although I already has some introductory experience and familiarity with the Godot Engine and its GDScript language, I identified a need to become more confident with its development environment and applying learned programming concepts with GDScript to complete a small project.

My PLP proposed the studying of the Godot documentation, watching, following along with, and completing tutorials, and undertaking practical exercises before going onto developing a small beginner game. These activities were selected as they provide a structured way to progress from my current basic understanding of the basics of the engine to a more in-depth understanding and broader skillset and developing a small game would provide me with the opportunity to apply the knowledge gained in a practical context. Rather than simply just following tutorials, I need to independently implement features into a game, troubleshoot any errors that occur, and consider how different elements of the game work together, allowing me to become more comfortable in applying my knowledge and working with GDScript and navigate the engine without following a tutorial one-to-one and requiring step-by-step guidance.

2.4. Goal 4 / Develop introductory skills within the C++ language

Included in my PLP was to develop introductory skills and familiarity with the C++ language, allowing me to broaden my programming knowledge, gaining experience with a language in which I have had no exposure to through my previous studies. As learning additional programming languages is a valuable skill within the computing industry, with different languages offering different approaches to solving problems and developing software, gaining C++ experience would allow me to strengthen my existing skills.

As I had no experience with C++, I identified this as an area of improvement in which I could develop new technical skills. My PLP proposed the use of beginner-level tutorials and completing programming exercises through platforms such as Leetcode, providing an accessible introduction, allowing me to develop a base level understanding of its syntax, structure, and programming concepts. Completing these exercises would also provide me with the opportunity to apply this knowledge practically, being required to attempt problems independently, interpreting the program requirements, developing my own appropriate solutions, and troubleshooting any errors I encounter during the implementation process, allowing me to become more comfortable in applying unfamiliar concepts and strengthen my ability to work independently whilst developing introductory C++ skills.

3. Evidence of Achievement

The following section summarises what each activity produced and the evidence demonstrating my progress towards completing the established goals. All supporting artefacts, including completion screenshots, documents, source code, and the completed, compiled game and its source repository, are available through the public portfolio linked on the title page and further referenced and indexed in Section 4.

3.1. Goal 1 / Demonstrate understanding of ethical principles and practices in a professional context, especially in relation to Indigenous Communities

I completed both the *Working Ethically in the IT Industry* and *Indigenous Professional* learning modules provided through the subjects Canvas. These activities allowed me to build upon the foundational understanding I had developed through my previous coursework, exploring professional and ethical responsibilities in greater depth. Through these activities, I was able to further explore the responsibilities of computing professionals and the importance of approaching development with both cultural awareness and respect. I also completed *Assessment 2: An Ethical Dilemma – Write to Your Manager*, which provided me with an opportunity to apply my understanding of ethical principles to a professional scenario. The completion of both modules and assignment provided an opportunity to develop my understanding of ethical principles, practices, and considerations within the IT industry particularly in relation to the importance of recognising cultural differences and considering the needs and perspectives of Indigenous communities. Supporting evidence, including module screenshots and the completed assignment are available on my portfolio.fi

3.2. Goal 2 / Demonstrate foundational skills in Cybersecurity

To achieve this goal, I completed the *Introduction to Cybersecurity* learning module provided through Canvas. This activity allowed me to build upon my existing knowledge base, providing me with a broader understanding of cybersecurity principles and practices. Through completing the module, I developed a greater understanding of common cybersecurity threats and vulnerabilities that can compromise computer systems and information and gained further insight into the importance of implementing security measures to reduce potential risk, prevent unauthorised access, and protect systems. To access my understanding, I completed both quizzes provided assessing the module, achieving a combined result of 40/40 (100%). These results provide evidence of my understanding of the material the module covers, demonstrating my progress towards achieving the goal and evidence of both assessment results are included in my professional development portfolio as supportive evidence.

3.3. Goal 3 / Develop foundational skills within the Godot Engine

To achieve this goal, I read through the official Godot documentation, followed and completed online tutorials, undertook practical exercises, and developed a small, beginner-level game allowing me to build upon my existing familarity with the engine, developing a better understanding of its environment and GDScript. Beginning with the official documentation and tutorials allowed me to familiarise myself with the engine's functionality and how GDScript is used before reinforcing this knowledge through

practical exercises and final completing my own beginner game, applying the programming concepts and knowledge I had learnt into practical experience. Completing the game provided me with a real, achievable goal point that demonstrated my ability to provide foundational skills, while also giving me practical experience and evidence of the skills developed through this activity. The completed game, its source code, and all other supportive evidence including tutorial and documentation screenshots are included in my portfolio.

3.4. Goal 4 / Develop introductory skills within the C++ language

Completing beginner-level C++ tutorials and programming exercises through the Leetcode platform allowed me to achieve this goal, developing an introductory understanding of the language. These activities allowed me to familiarise myself with the language's syntax, structure, and fundamental programming concepts, expanding my existing programming knowledge base. Starting with the following tutorials and completing quizzes provided me with a baseline introduction to C++ readying me to move onto completing regular exercises on LeetCode where I applied the learned programming concepts independently, providing practical experience in writing C++ code, troubleshooting, and developing problem solutions in the language. Through this I established a foundational knowledge base and introductory skills in C++. All supporting evidence, including screenshots and all relevant code, is included in my portfolio.

4. Integration with the PLP and Portfolio

The report, portfolio, and original PLP are intended to be read together as a record documenting my professional learning throughout the semester, with the PLP establishing four goals while the portfolio provides supporting evidence of their completion. The following table maps each PLP goal to its corresponding portfolio section and supporting artefacts.

PLP Goal	Portfolio Section	Key Artefacts
Goal 1 / Ethical principles and practices.	Ethics	Working Ethically in the IT Industry module; Indigenous Professional Capability module; Assessment 2: An Ethical Dilemma, Write to Your Manager; completion screenshots
Goal 2 / Cybersecurity	Cybersecurity	Introduction to Cybersecurity module; Assessment 1 and Assessment 2 quiz results (40/40 combined); completion screenshots
Goal 3 / Godot Engine	Godot	Official Godot documentation; completed online tutorials and

		exercises; GDScript code; completed beginner game and source code; screenshots of development progress.
Goal 4 / C++	C++	Completed C++ tutorials; completed LeetCode exercises; relevant C++ source code.

5. Reflection and Analysis

Reflecting upon my professional development throughout the semester, completing the activities outlined in my PLP has provided me with valuable opportunities to broaden my existing knowledge, addressing gaps in my knowledge. At the beginning of the semester, I identified these gaps and established goals intended to develop both my technical capabilities in the computing industry and professional awareness. Through these activities I have developed a deeper understanding of the knowledgebase required within the computing industry, extending far beyond just programming and technical knowledge, which although still extremely important, must also encompass ethical responsibility such as cultural awareness and protecting users' privacy and personal information through cybersecurity. Though I still recognise the importance of the former, as I gained practical experience too with the unfamiliar technologies of Godot and C++, allowing to expand my programming knowledge base and explore beyond my coursework.

Developing my understanding of both ethical principles and practices and the area of cybersecurity provided me with valuable insight into the broader responsibilities of those in the computing industry. Though my previous coursework provided me with a baseline understanding of ethical considerations, particularly through my project on Saibai Island and its Indigenous community, completing the provided modules greatly expanded upon this understanding, reinforcing the importance of considering the cultural perspective of the Indigenous communities' technological solutions may affect. Similarly, the cybersecurity module complimented my existing knowledge of networking and Unix systems, introducing me to the threats commonly faced by systems and how they are projected against such potential vulnerabilities. Together, these both reinforced that to develop reliable technological solutions you must heavily consider the responsibilities and ethicality of the solution alongside any technical implementation, better preparing me for a career in the computing industry.

My technical development goals both provided me with the opportunity to explore new, unfamiliar technologies, broadening my programming experience, better preparing me for a career in the computing industry. Through studying the Godot documentation, completing and following alongside tutorials, completing practical exercises, and

developing a small beginner game, I gained a sense of familiarity with the engine and developed foundational skills within its GDScript language. A core takeaway from this was taking and understanding programming concepts learned through a guided context and being able to apply them independently through a practical exercise. Similarly, learning the basics of C++ introduced me to an entirely unfamiliar programming language with a different syntax and programming concepts which I was required to develop an understanding of before applying them independently through programming exercises. Both learning experiences reinforced the importance of practical application when learning something new as simply becoming familiar with something does not necessarily translate into proficiency or experience.

The most important lesson I have taken from this experience is the value of independent learning above all, as through these activities I aimed to learn outside of the preassigned coursework, relying instead of existing documentations, tutorials, and online learning resources, before applying this through practical exercises. This reinforced the importance of taking the initiative when exploring unfamiliar areas and made me recognise that learning isn't only confined to formal education, allowing me to become more comfortable seeking information independently.

Overall, the completion of these activities this semester allowed me to make meaningful progress towards the goals the PLP established for me while still developing a clearer understanding of my existing capabilities and areas in which I still have room for further improvement. I recognise that despite all the skills I have developed through this course and how much my knowledge base has expanded, they are still only a foundational understanding and provide a baseline for continued further development. Therefore, building upon this experience, I have outlined future development goals in the following section, providing how I intend to keep improving my professional capabilities and better prepare me for a career in the computing industry.

6. Goals for future professional development

Reflecting on my professional development, the three goals outlined below have been selected to build upon the skills and knowledge I have cultivated throughout the semester. Each focus on a specific area for further improvement, outlining achievable objectives providing opportunities to continue my professional development. The first aims to further develop my existing game development skills through the completion of a larger, more substantial project; the second on strengthening my C++ skills and proficiency; and the third explores new creative skills important to game development and complement my existing technical skillset.

6.1. Goal 1 / Further Develop Game Development Skills Using Godot

Having established a foundational understanding of the Godot engine and its GDScript language throughout the semester, I wish to continue to develop my game development skillset through the undertaking of a larger, more substantial project. Whilst my initial professional learning goal adequately provided me with a foundational knowledge base, familarising me with the engine and the application of basic game development concepts, I believe the natural progression to this is to challenge myself to develop a more complete game with greater functionality and complexity.

To achieve this, my intention is to undertake a personal game development project utilising the knowledge I have gained throughout my previous learning activities as a starting point. Rather than relying heavily on tutorials, I would like to independently research unfamiliar topics or when I'm stuck on something, allowing me to develop a greater understanding, strengthening my ability to plan and structure projects, trouble shoot technical issues, and independently solve problems while exploring advanced features in the engine.

6.2. Goal 2 / Further Develop Programming Skills and Proficiency in C++

Building upon the introductory skills developed throughout this semester, I would like to further develop my proficiency in the language, broadening my programming capabilities. While my initial goal focused on learning syntax and fundamental concepts, I recognise the need for further practice, allowing me to be able to apply these skills both confidently and independently. To achieve this, I intend to continue completing programming exercises, however, gradually progress towards more challenging problems as well as branching out into exploring more advanced concepts and undertaking small personal projects which would allow me to apply my knowledge in a practical context.

6.3. Goal 3 / Develop Skills in Game Sprite and Asset Creation

Alongside developing my game development skills, I would like to explore the creative aspects of game development and sprite creation. While my previous goal primarily focused on the technical aspects of game development, I recognise that visual design plays a highly important role in game development and developing these skills would allow me to broaden my game development skillset and my ability to create more comprehensive games. To achieve this, I intend to explore digital art tools and follow tutorials covering areas such as pixel art, sprite design, and basic animation, and apply these concepts by practicing creating original assets to gain practical experience.

References

Australian Computer Society. (2023). *Code of Professional Ethics*. https://www.acs.org.au/governance/rules-and-regulations.html

Australian Institute of Aboriginal and Torres Strait Islander Studies. (2020). *AIATSIS code of ethics for Aboriginal and Torres Strait Islander Research.* *https://aiatsis.gov.au/research/ethical-research/code-ethics*